

Power BI Assignment 2 - DAX & Data Visualization Report

1. Project Overview

This report documents the Power BI Assignment 2 project, including DAX calculations, interactive dashboards, data analysis, and business insights generated from the E-Commerce Sales dataset.

2. DAX Formulas Used

Insert screenshots of the following DAX formulas here:

- Category Type (Calculated Column)
- Revenue per Order (Calculated Column)
- Sales Category (Calculated Column)
- Order Count (Measure)
- Average Profit in Delhi (Measure)
- YTD Sales (Measure)

The screenshot shows the Power BI DAX editor interface. The top section displays the formula for the 'Category Type' calculated column: `1 Category Type =`
`2 'Order Details'[Category] & " - " & 'Order Details'[Sub-Category]`

The bottom section displays the formula for the 'Revenue per Order' calculated column: `1 Revenue per Order =`
`2 'Order Details'[Amount] * 'Order Details'[Quantity]`

Below the formulas, two screenshots of the data table are shown. The first screenshot shows the 'Order Details' table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, and Category Type. The second screenshot shows the 'Order Details' table with columns: Order ID, Amount, Profit, Quantity, Category, Sub-Category, Category Type, and Revenue per Order.

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type
B-25601	₹ 8	2 - ₹	3	Clothing	Hankerchief	Clothing - Hankerchief
B-25602	₹ 561	₹ 212	3	Clothing	Saree	Clothing - Saree
B-25603	₹ 193	166 - ₹	3	Clothing	Saree	Clothing - Saree
B-25603	₹ 180	₹ 5	3	Clothing	Trousers	Clothing - Trousers
B-25621	₹ 228	₹ 63	3	Electronics	Electronic Games	Electronics - Electronic Games
B-25621	₹ 333	15 - ₹	3	Electronics	Phones	Electronics - Phones
B-25623	₹ 158	₹ 69	3	Clothing	Stole	Clothing - Stole
B-25625	₹ 59	₹ 30	3	Clothing	Leggings	Clothing - Leggings
B-25626	₹ 1,103	276 - ₹	3	Furniture	Chairs	Furniture - Chairs
B-25629	₹ 1,560	₹ 421	3	Clothing	Trousers	Clothing - Trousers
B-25630	₹ 40	7 - ₹	3	Clothing	Stole	Clothing - Stole
B-25634	₹ 389	83 - ₹	3	Furniture	Chairs	Furniture - Chairs
B-25637	₹ 117	₹ 14	3	Clothing	Shirt	Clothing - Shirt
B-25638	₹ 182	11 - ₹	3	Furniture	Bookcases	Furniture - Bookcases
B-25638	₹ 154	₹ 39	3	Clothing	Hankerchief	Clothing - Hankerchief
B-25638	₹ 816	96 - ₹	3	Electronics	Printers	Electronics - Printers
B-25639	₹ 1,629	153 - ₹	3	Electronics	Phones	Electronics - Phones
B-25650	₹ 269	₹ 111	3	Clothing	Trousers	Clothing - Trousers
B-25651	₹ 172	103 - ₹	3	Furniture	Chairs	Furniture - Chairs

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order
B-25601	₹ 8	2 - ₹	3	Clothing	Hankerchief	Clothing - Hankerchief	₹ 24
B-25602	₹ 561	₹ 212	3	Clothing	Saree	Clothing - Saree	₹ 1,683
B-25603	₹ 193	166 - ₹	3	Clothing	Saree	Clothing - Saree	₹ 579
B-25603	₹ 180	₹ 5	3	Clothing	Trousers	Clothing - Trousers	₹ 540
B-25621	₹ 228	₹ 63	3	Electronics	Electronic Games	Electronics - Electronic Games	₹ 684
B-25621	₹ 333	15 - ₹	3	Electronics	Phones	Electronics - Phones	₹ 999
B-25623	₹ 158	₹ 69	3	Clothing	Stole	Clothing - Stole	₹ 474
B-25625	₹ 59	₹ 30	3	Clothing	Leggings	Clothing - Leggings	₹ 177
B-25626	₹ 1,103	276 - ₹	3	Furniture	Chairs	Furniture - Chairs	₹ 3,309
B-25629	₹ 1,560	₹ 421	3	Clothing	Trousers	Clothing - Trousers	₹ 4,680
B-25630	₹ 40	7 - ₹	3	Clothing	Stole	Clothing - Stole	₹ 120
B-25634	₹ 389	83 - ₹	3	Furniture	Chairs	Furniture - Chairs	₹ 1,167
B-25637	₹ 117	₹ 14	3	Clothing	Shirt	Clothing - Shirt	₹ 351
B-25638	₹ 182	11 - ₹	3	Furniture	Bookcases	Furniture - Bookcases	₹ 546
B-25638	₹ 154	₹ 39	3	Clothing	Hankerchief	Clothing - Hankerchief	₹ 462
B-25638	₹ 816	96 - ₹	3	Electronics	Printers	Electronics - Printers	₹ 2,448
B-25639	₹ 1,629	153 - ₹	3	Electronics	Phones	Electronics - Phones	₹ 4,887
B-25650	₹ 269	₹ 111	3	Clothing	Trousers	Clothing - Trousers	₹ 807
B-25651	₹ 172	103 - ₹	3	Furniture	Chairs	Furniture - Chairs	₹ 516

Manage relationships

New Quick New New

measure measure column table

Calculations

Mark as date table

Calendars

Structure

1 Sales Category =

2 IF(

3 'Order Details'[Amount] > AVERAGE('Order Details'[Amount]),

4 "Above Average",

5 "Below Average",

6)

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
B-25601	₹ 8	2 -₹	3	Clothing	Hankерchief	Clothing - Hankерchief	₹ 24	Below Average
B-25602	₹ 561	₹ 212	3	Clothing	Saree	Clothing - Saree	₹ 1,683	Below Average
B-25603	₹ 193	166 -₹	3	Clothing	Saree	Clothing - Saree	₹ 579	Below Average
B-25603	₹ 180	₹ 5	3	Clothing	Trousers	Clothing - Trousers	₹ 540	Below Average
B-25621	₹ 228	₹ 63	3	Electronics	Electronic Games	Electronics - Electronic Games	₹ 684	Below Average
B-25621	₹ 333	15 -₹	3	Electronics	Phones	Electronics - Phones	₹ 999	Below Average
B-25623	₹ 158	₹ 69	3	Clothing	Stole	Clothing - Stole	₹ 474	Below Average
B-25625	₹ 59	₹ 30	3	Clothing	Leggings	Clothing - Leggings	₹ 177	Below Average
B-25626	₹ 1,103	276 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 3,309	Above Average
B-25629	₹ 1,560	₹ 421	3	Clothing	Trousers	Clothing - Trousers	₹ 4,680	Above Average
B-25630	₹ 40	7 -₹	3	Clothing	Stole	Clothing - Stole	₹ 120	Below Average
B-25634	₹ 389	83 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 1,167	Below Average
B-25637	₹ 117	₹ 14	3	Clothing	Shirt	Clothing - Shirt	₹ 351	Below Average
B-25638	₹ 182	11 -₹	3	Furniture	Bookcases	Furniture - Bookcases	₹ 546	Below Average
B-25638	₹ 154	₹ 39	3	Clothing	Hankерchief	Clothing - Hankерchief	₹ 462	Below Average
B-25638	₹ 816	96 -₹	3	Electronics	Printers	Electronics - Printers	₹ 2,448	Above Average

Data

Q Search

> Average profit

> List of Orders

> Monthly Target

> Order count

> Order Details

Σ Amount

Category

Category Type

Order ID

Σ Profit

Σ Quantity

Revenue per Order

Category Sales Category

Sub-Category

> Orders Data

> Sales target

Structure

1 Order count =

2 COUNT('Order Details'[Order ID])

Order ID	Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category
B-25601	₹ 8	2 -₹	3	Clothing	Hankерchief	Clothing - Hankерchief	₹ 24	Below Average
B-25602	₹ 561	₹ 212	3	Clothing	Saree	Clothing - Saree	₹ 1,683	Below Average
B-25603	₹ 193	166 -₹	3	Clothing	Saree	Clothing - Saree	₹ 579	Below Average
B-25603	₹ 180	₹ 5	3	Clothing	Trousers	Clothing - Trousers	₹ 540	Below Average
B-25621	₹ 228	₹ 63	3	Electronics	Electronic Games	Electronics - Electronic Games	₹ 684	Below Average
B-25621	₹ 333	15 -₹	3	Electronics	Phones	Electronics - Phones	₹ 999	Below Average
B-25623	₹ 158	₹ 69	3	Clothing	Stole	Clothing - Stole	₹ 474	Below Average
B-25625	₹ 59	₹ 30	3	Clothing	Leggings	Clothing - Leggings	₹ 177	Below Average
B-25626	₹ 1,103	276 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 3,309	Above Average
B-25629	₹ 1,560	₹ 421	3	Clothing	Trousers	Clothing - Trousers	₹ 4,680	Above Average
B-25630	₹ 40	7 -₹	3	Clothing	Stole	Clothing - Stole	₹ 120	Below Average
B-25634	₹ 389	83 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 1,167	Below Average
B-25637	₹ 117	₹ 14	3	Clothing	Shirt	Clothing - Shirt	₹ 351	Below Average
B-25638	₹ 182	11 -₹	3	Furniture	Bookcases	Furniture - Bookcases	₹ 546	Below Average
B-25638	₹ 154	₹ 39	3	Clothing	Hankерchief	Clothing - Hankерchief	₹ 462	Below Average
B-25638	₹ 816	96 -₹	3	Electronics	Printers	Electronics - Printers	₹ 2,448	Above Average
B-25639	₹ 1,629	153 -₹	3	Electronics	Phones	Electronics - Phones	₹ 4,887	Above Average
B-25650	₹ 269	₹ 111	3	Clothing	Trousers	Clothing - Trousers	₹ 807	Below Average
B-25651	₹ 172	103 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 516	Below Average
B-25652	₹ 174	70 -₹	3	Electronics	Accessories	Electronics - Accessories	₹ 522	Below Average

Data

Q Search

> Average profit

> List of Orders

> Monthly Target

> Order count

> Order Details

Σ Amount

Category

Category Type

Order Count

Order ID

Σ Profit

Σ Quantity

Revenue per Order

Sales Category

Sub-Category

> Orders Data

> Sales target

1 Average Profit in Delhi = 2 CALCULATE(3 AVERAGE('Order Details'[Profit]), 4 'List of Orders'[City] = "Delhi" 5)								
Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category	
₹ 8	2 -₹	3	Clothing	Hankerchief	Clothing - Hankerchief	₹ 24	Below Average	
₹ 561	₹ 212	3	Clothing	Saree	Clothing - Saree	₹ 1,683	Below Average	
₹ 193	166 -₹	3	Clothing	Saree	Clothing - Saree	₹ 579	Below Average	
₹ 180	₹ 5	3	Clothing	Trousers	Clothing - Trousers	₹ 540	Below Average	
₹ 228	₹ 63	3	Electronics	Electronic Games	Electronics - Electronic Games	₹ 684	Below Average	
₹ 333	15 -₹	3	Electronics	Phones	Electronics - Phones	₹ 999	Below Average	Name (Average Profit)
₹ 158	₹ 69	3	Clothing	Stole	Clothing - Stole	₹ 474	Below Average	
₹ 59	₹ 30	3	Clothing	Leggings	Clothing - Leggings	₹ 177	Below Average	
₹ 1,103	276 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 3,309	Above Average	
₹ 1,560	₹ 421	3	Clothing	Trousers	Clothing - Trousers	₹ 4,680	Above Average	
₹ 40	7 -₹	3	Clothing	Stole	Clothing - Stole	₹ 120	Below Average	
₹ 389	83 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 1,167	Below Average	
₹ 117	₹ 14	3	Clothing	Shirt	Clothing - Shirt	₹ 351	Below Average	
₹ 182	11 -₹	3	Furniture	Bookcases	Furniture - Bookcases	₹ 546	Below Average	
₹ 154	₹ 39	3	Clothing	Hankerchief	Clothing - Hankerchief	₹ 462	Below Average	
₹ 816	96 -₹	3	Electronics	Printers	Electronics - Printers	₹ 2,448	Above Average	
₹ 1,629	153 -₹	3	Electronics	Phones	Electronics - Phones	₹ 4,887	Above Average	
₹ 260	₹ 111	3	Clothing	Trousers	Clothing - Trousers	₹ 807	Below Average	

Structure

Formatting

Properties

Calculations

1 YTD Sales = 2 TOTALYTD(3 SUM('Order Details'[Amount]), 4 'List of Orders'[Order Date] 5)								
Amount	Profit	Quantity	Category	Sub-Category	Category Type	Revenue per Order	Sales Category	
₹ 8	2 -₹	3	Clothing	Hankerchief	Clothing - Hankerchief	₹ 24	Below Average	
₹ 561	₹ 212	3	Clothing	Saree	Clothing - Saree	₹ 1,683	Below Average	
₹ 193	166 -₹	3	Clothing	Saree	Clothing - Saree	₹ 579	Below Average	
₹ 180	₹ 5	3	Clothing	Trousers	Clothing - Trousers	₹ 540	Below Average	
₹ 228	₹ 63	3	Electronics	Electronic Games	Electronics - Electronic Games	₹ 684	Below Average	
₹ 333	15 -₹	3	Electronics	Phones	Electronics - Phones	₹ 999	Below Average	
₹ 158	₹ 69	3	Clothing	Stole	Clothing - Stole	₹ 474	Below Average	
₹ 59	₹ 30	3	Clothing	Leggings	Clothing - Leggings	₹ 177	Below Average	
₹ 1,103	276 -₹	3	Furniture	Chairs	Furniture - Chairs	₹ 3,309	Above Average	
₹ 1,560	₹ 421	3	Clothing	Trousers	Clothing - Trousers	₹ 4,680	Above Average	

3. Created Visualizations

Insert screenshots of each visualization and add a short explanation below each one.

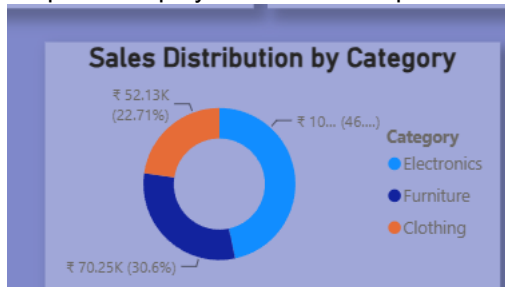
- **Clustered Column Chart**

Purpose: Compares Sales Target Achievement across different product categories.



• Donut Chart

Purpose: Displays the maximum profit margin by sub-category for easy comparison.



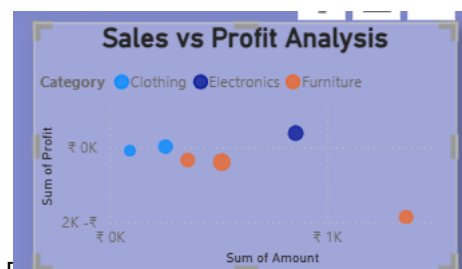
• Line Chart

Purpose: Shows the monthly sales trend and helps identify seasonal patterns.



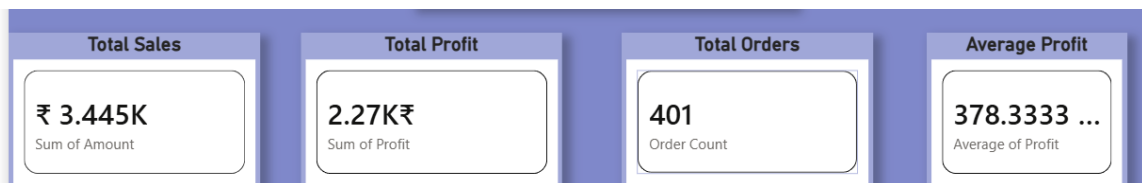
• Scatter Chart

Purpose: Analyzes the relationship between Profit and Quantity by sub-category.



• KPI Cards

Purpose: Displays key business metrics such as Total Sales, Profit, Orders and Sales Target. [



• Matrix

Purpose: Provides a detailed sales summary grouped by Category and Sub-Category. [

Category	Bookcases	Chairs	Hankerchief	Printers	Saree	Tables	Total
Furniture	₹ 359	₹ 516				₹ 1,364	₹ 2,239
Electronics				₹ 856			₹ 856
Clothing			₹ 257		₹ 93		₹ 350
Total	₹ 359	₹ 516	₹ 257	₹ 856	₹ 93	₹ 1,364	₹ 3,445

Sales Matrix by Category & Sub-Category

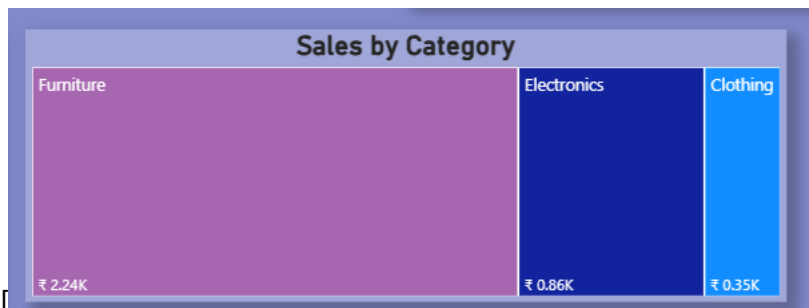
• Map

Purpose: Visualizes sales distribution across cities and states.



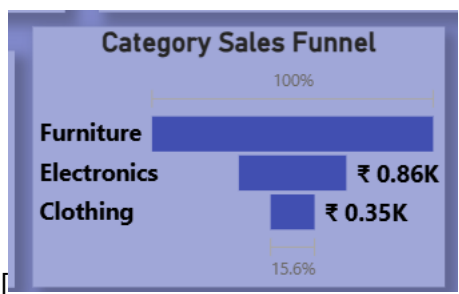
• Treemap

Purpose: Displays sales contribution by sub-category using proportional rectangles.



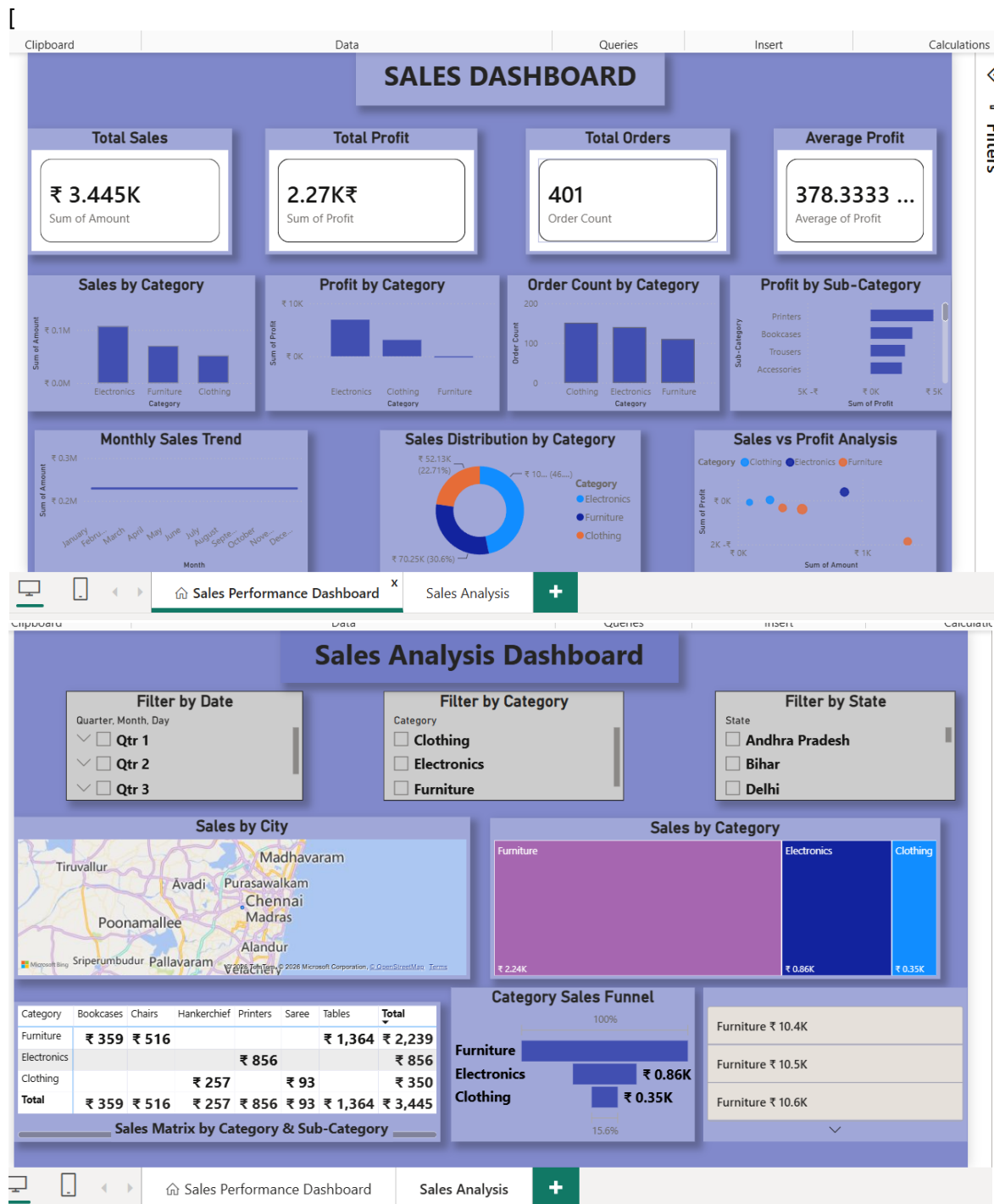
• Funnel Chart

Purpose: Compares order count across states and highlights top-performing regions.



4. Dashboard Layout

Insert a full-page screenshot of the Dashboard and another screenshot of the Analysis page.



5. Conclusion

The project demonstrates practical knowledge of Power BI, DAX, data visualization, dashboard development, and business intelligence reporting.